Top 50 AI Engineer Answers

1. What is AI? How is it different from ML and DL?

AI is the broader field of creating intelligent systems. ML is a subset of AI that learns from data, while DL is a subset of ML that uses neural networks with multiple layers.

2. What is Machine Learning? Explain its types.

Machine Learning enables systems to learn patterns from data without explicit programming. Its main types are supervised, unsupervised, and reinforcement learning.

3. What is Deep Learning? How is it different from ML?

Deep Learning uses multi-layer neural networks to automatically learn features from data. Traditional ML often requires manual feature engineering.

4. Difference between supervised, unsupervised and reinforcement learning?

Supervised learning uses labeled data, unsupervised learning finds hidden patterns in unlabeled data, and reinforcement learning learns through rewards and penalties.

5. What is overfitting and how do you prevent it?

Overfitting happens when a model memorizes training data and performs poorly on new data. It can be reduced using regularization, dropout, and more training data.

6. What is underfitting and how do you fix it?

Underfitting occurs when a model is too simple to learn patterns. Increasing model complexity or training longer can help.

7. What is bias-variance tradeoff?

Bias is error due to overly simple assumptions, while variance is sensitivity to training data. A good model balances both.

8. Difference between classification and regression?

Classification predicts categories or labels, while regression predicts continuous numerical values.

9. Evaluation metrics for classification?

Common metrics include Accuracy, Precision, Recall, F1-Score, and ROC-AUC depending on the problem.

10. Evaluation metrics for regression?

MAE, MSE, RMSE, and R^2 are commonly used to measure regression performance.

11. What is cross-validation?

Cross-validation splits data into multiple folds to evaluate model performance more reliably and reduce overfitting risk.

12. What is feature engineering?

Feature engineering is creating or transforming input features to improve model performance and learning.

13. How do you handle missing values?

Missing values can be removed, filled using mean/median/mode, or predicted using advanced imputation techniques.

14. What is normalization and standardization?

Normalization scales values to a fixed range, while standardization transforms data to have zero mean and unit variance.

15. What is the curse of dimensionality?

As features increase, data becomes sparse and models require more data and computation to learn effectively.

16. What is PCA?

Principal Component Analysis reduces dimensionality by transforming features into fewer components while retaining maximum variance.

17. Difference between L1 and L2 regularization?

L1 can reduce some weights to zero for feature selection, while L2 shrinks weights smoothly to reduce overfitting.

18. What is a confusion matrix?

A confusion matrix summarizes classification results using True Positives, True Negatives, False Positives, and False Negatives.

19. Difference between precision, recall and F1-score?

Precision measures correctness of positive predictions, recall measures how many positives were found, and F1 balances both.

20. What is ROC-AUC?

ROC-AUC measures a model's ability to distinguish between classes across different classification thresholds.

21. Difference between batch, mini-batch and SGD?

Batch uses the entire dataset, SGD updates after each sample, and mini-batch updates using small groups of samples.

22. What is a neural network?

A neural network consists of interconnected neurons organized into layers that learn complex patterns from data.

23. What is an activation function?

Activation functions introduce non-linearity, allowing neural networks to learn complex relationships. Examples include ReLU and Sigmoid.

24. What is the vanishing gradient problem?

Gradients become extremely small during backpropagation, making deep networks difficult to train effectively.

25. What is the exploding gradient problem?

Gradients become excessively large, causing unstable training and poor convergence.

26. Purpose of a loss function?

A loss function measures prediction error and guides the model during optimization.

27. What is backpropagation?

Backpropagation calculates gradients of the loss and updates network weights using gradient descent.

28. Difference between CNN, RNN and Transformers?

CNNs are effective for images, RNNs handle sequences, and Transformers use attention mechanisms for parallel processing.

29. What is a CNN?

A Convolutional Neural Network extracts spatial features from images using convolution and pooling layers.

30. What is an RNN?

RNNs process sequential data by maintaining information from previous time steps but struggle with long-term dependencies.

31. What is LSTM?

LSTM is a special RNN architecture that uses gates to remember important information over long sequences.

32. What is a Transformer?

Transformers use self-attention to understand relationships between tokens and are the foundation of modern LLMs.

33. Difference between training, validation and test sets?

Training data trains the model, validation data tunes hyperparameters, and test data evaluates final performance.

34. How do you choose a machine learning model?

Consider data size, problem type, interpretability, computational cost, and performance requirements.

35. What is hyperparameter tuning?

It is the process of finding optimal parameter values using Grid Search, Random Search, or Bayesian Optimization.

36. What is an embedding?

Embeddings are dense vector representations that capture semantic meaning and relationships between items.

37. What is transfer learning?

Transfer learning reuses knowledge from a pre-trained model to solve a related task with less data.

38. What is fine-tuning?

Fine-tuning adapts a pre-trained model to a specific task by continuing training on task-specific data.

39. What is Generative AI?

Generative AI creates new content such as text, images, audio, or code based on learned patterns.

40. Difference between discriminative and generative models?

Discriminative models classify data, while generative models learn data distribution and generate new samples.

41. What are Large Language Models?

LLMs are transformer-based models trained on massive text datasets to understand and generate human-like language.

42. How do LLMs work at a high level?

They predict the next token in a sequence using patterns learned during large-scale training.

43. What is prompt engineering?

Prompt engineering is designing effective instructions that guide LLMs to produce desired outputs.

44. Best practices for prompt engineering?

Provide clear instructions, context, examples, constraints, and specify the expected output format.

45. Difference between zero-shot, one-shot and few-shot prompting?

Zero-shot uses no examples, one-shot uses one example, and few-shot uses multiple examples in the prompt.

46. What is RAG?

Retrieval-Augmented Generation combines external knowledge retrieval with LLM generation for more accurate responses.

47. What are AI Agents?

AI agents are systems that perceive, reason, and take actions autonomously to achieve goals.

48. What is Agentic AI?

Agentic AI refers to AI systems capable of planning, tool usage, decision-making, and autonomous execution.

49. Key components of an AI Agent?

Typical components include an LLM, memory, tools, planning module, and execution framework.

50. How do you build an Agentic AI system?

Define goals, connect tools and memory, implement planning logic, and continuously evaluate performance.